QP CODE: 25901745

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCPEG100 - DRILLING, TRANSPORTATION ANDREFINING TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes

Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest (1) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple choice questions

Answer all questions

Each question carries 1 mark [1x10 = 10]

1. How can water circulation contribute to scaling in pipelines? [K]/[CO1]

c). By depositing calcium carbonate and sulfate compounds

2. The presence of sulfate-reducing bacteria (SRB) in oil field water leads to: [U]/[CO1]

d). Hydrogen sulfide production

3. What is the main purpose of secondary oil recovery methods like gas flooding and water flooding? [U]/[CO1]

d). To increase the pressure in the reservoir and extract more oil after primary recovery.

4. What is an advantage of GTL fuels over traditional petroleum fuels? [U]/[C01]

c). Lower emissions and cleaner burning

5. A company wants to produce a wide range of refined products, including high-value petrochemicals and specialized fuels. Which type of refinery should they choose? [U]/[CO3]

c). Integrated refinery

6. Which distillation method would you recommend for separating a mixture of compounds with boiling points differing by less than 5°C? [E]/[CO4]

a). Fractional distillation

7. Identify the heat exchanger type that uses concentric tube construction: [K]/[CO5]

c). Double-pipe heat exchanger

8. High concentrations of calcium and sulfate in brines can lead to the formation of: [U]/[CO1]

c). Gypsum scaling

9. Specific gravity is the ratio of the density of a substance to the density of. [K]/[CO6]

c). Water

10. The standard method used to determine the cloud point: [U]/[CO6]
c). ASTM D2500

Part B

Short Answer Questions

Answer any 4 questions

Each question carries 3 marks **[3x4 = 12]**

11. Explain the dissolved gas drive in primary oil recovery. [U]/[CO1]
In this method, natural gas dissolved in the reservoir oil helps drive the oil to the surface when reservoir pressure is released. When a well is drilled, the pressure drop causes the dissolved gas to expand, pushing the oil toward the wellbore. This method is effective in reservoirs with high pressure and gas content, but its efficiency decreases over time as reservoir pressure declines.

(Any three points 3 Marks)

12. List out the safety consideration of nitrogen blanketing. [K]/[CO2]
Regular oxygen monitoring should be carried out to detect leaks and maintain nitrogen purity inside the tank. Proper ventilation around the tank is essential to prevent nitrogen accumulation, which may cause asphyxiation hazards in confined spaces. In addition, nitrogen pressure must be carefully controlled to avoid tank rupture or product loss due to excessive pressure.

(Any three points 3 Marks)

13. Describe the composition of fuel oil and its applications. [K]/[CO4]
Fuel oil is a heavy petroleum fraction mainly composed of long-chain hydrocarbons along with small amounts of sulfur, nitrogen, oxygen compounds, and impurities. It is obtained during the refining of crude oil after lighter products like gasoline and diesel are separated. Fuel oil is widely used in industrial boilers, furnaces, power plants, marine engines, and heating systems due to its high calorific value. **(Composition 1.5 Marks Any three applications 1.5 Marks)**

14. What is Natural Rubber Modified Bitumen (NRMB)? [K]/[CO4]
Natural Rubber Modified Bitumen (NRMB) is a type of modified bitumen (asphalt) where natural rubber is added to enhance the properties of the base bitumen. This modification improves the performance and durability of the bitumen, particularly in road construction and paving applications. **(Any three points 3 Marks)**

15. Discuss the main functions of drilling mud in oil and gas drilling. [U]/[CO1]
Drilling mud, also known as drilling fluid- a fluid circulated through the wellbore to cool the drill bit, remove cuttings, and maintain pressure control. The main functions of drilling mud in oil and gas drilling is cooling and lubricating the drill bit to reduce wear and tear, carrying cuttings to the surface to prevent clogging of the wellbore and also controlling formation pressure to prevent blowouts. **(Any three points 3 Marks)**

16. Explain the principle of the Abel flash point test. [U]/[CO6]
The Abel flash point test is used to determine the lowest temperature at which petroleum products give off enough vapour to ignite momentarily in the presence of a flame. In this method, the oil sample is heated slowly in a closed cup apparatus, and a small test flame is introduced at regular

temperature intervals. The temperature at which a brief flash appears is recorded as the flash point of the oil.

Part C

Short Essay Questions

Answer any 4 questions

Each question carries 7 marks **[7x4 = 28]**

17. Discuss the main safety concerns on oil rigs, and how can they be reduced. [U]/[CO1]

The main safety concerns on oil rigs include equipment failures, fire and explosion risks, personnel safety issues, and harsh weather conditions. Since oil rigs operate in hazardous environments, proper safety measures are essential to prevent accidents and protect workers.

1. Equipment Failures

Critical equipment such as Blowout Preventers (BOPs) and pressure control systems are essential for safe drilling operations. Failure of these systems can cause uncontrolled release of oil and gas, leading to blowouts and major accidents. Structural components of rigs may also fail due to corrosion, strong winds, and high waves. These risks can be reduced through regular maintenance, inspections, and real-time monitoring of equipment.

2. Fire and Explosion Risks

Oil rigs handle flammable gases, crude oil, and chemicals, making them highly vulnerable to fires and explosions. Sparks, gas leaks, or equipment malfunctions may trigger accidents. Installing fire detection and suppression systems, ensuring proper ventilation, and conducting regular emergency and fire drills help reduce these hazards.

3. Personnel Safety

Workers on rigs face risks from heavy machinery, slippery surfaces, confined spaces, and elevated platforms. Human error, fatigue, and lack of training can also cause accidents. These risks can be minimized by providing proper safety training, personal protective equipment (PPE), fall protection systems, and safe working procedures.

4. Weather Conditions

Offshore rigs are exposed to severe weather conditions such as storms, strong winds, rough seas, and freezing temperatures, which can make operations dangerous. Safety can be improved by implementing emergency response protocols, evacuation plans, and using strong rig structures designed to withstand harsh environments. **(1 mark for the main headings of safety concern & Any two points under one sub heading give 1.5 Marks each)**

18. Compare and contrast the environmental risks of transporting oil and gas by pipeline versus by tanker (ship). [U]/[CO2]

Pipelines are commonly used for transporting oil and gas over long distances on land and underwater. Pipelines are usually safer and more energy-efficient for continuous transport, but leaks may go unnoticed for long periods and affect land ecosystems. Pipeline accidents are generally localized. The main environmental risk is leakage caused by corrosion, equipment failure, or accidental damage. Pipeline leaks can contaminate soil, groundwater, rivers, and agricultural land. Gas pipeline leaks may also release methane, a harmful greenhouse gas.

However, pipelines generally have lower accident rates and produce less air pollution during normal operation.

Tankers are mainly used for transporting crude oil and liquefied natural gas across oceans. Tankers can transport large quantities internationally, but accidents may cause sudden and catastrophic marine pollution. The major environmental risk is oil spills caused by collisions, grounding, storms, or equipment failure. Large oil spills can severely damage marine ecosystems, kill fish and birds, and pollute coastlines for many years. Tankers also contribute to air pollution through fuel emissions and may introduce invasive species through ballast water discharge. Tanker spills can spread over vast ocean areas and have long-lasting environmental effects.

(3.5 marks each for Pipeline and Tankers)

19. Explain the composition, functions and properties of Light Diesel Oil and Lubricating oil. [U]/[CO4]

Light Diesel Oil (LDO): Composition: Light Diesel Oil is a middle distillate petroleum product obtained during crude oil refining. It mainly contains hydrocarbons such as paraffins, naphthenes, and aromatic compounds with carbon numbers generally ranging from C10 to C20.

Functions

- Used as fuel in diesel engines, boilers, furnaces, and generators.
- Provides energy for industrial heating and power generation.
- Used in agricultural pumps and heavy machinery.

Properties

- Moderate viscosity and good flow characteristics.
- High calorific value and good combustion efficiency.
- Higher flash point than petrol, making it safer to handle.
- Low volatility and good storage stability.

Lubricating Oil: Composition: Lubricating oil is produced from heavy fractions of crude oil and contains high molecular weight hydrocarbons along with additives such as antioxidants, detergents, anti-wear agents, and corrosion inhibitors.

Functions

- Reduces friction and wear between moving machine parts.
- Helps in cooling engine components by removing heat.
- Prevents corrosion and rust formation.
- Cleans engine parts by carrying away dirt and metal particles.
- Provides sealing between moving parts.

Properties

- High viscosity and good viscosity index.
- High boiling point and low volatility.
- Good thermal and oxidation stability.
- Non-corrosive and resistant to foaming.
- Good lubricating ability under high temperature and pressure conditions.

(1.5 mark for Composition, 1 mark each for any two points under functions & properties for LDO & Lubricating oil)

20. Discuss the differences between Aromatic Naphtha (HAN) and Paraffinic Naphtha (LAN) in terms of composition and applications. [U]/[CO4]

Aromatic Naphtha (HAN) and Paraffinic Naphtha (LAN) are petroleum-derived solvents that differ mainly in their chemical composition and industrial applications.

Heavy Aromatic Naphtha (HAN) contains a high percentage of aromatic hydrocarbons such as benzene, toluene, and xylene. These compounds have ring-shaped molecular structures and higher solvency power.

Applications

Used as an industrial solvent in paints, coatings, varnishes, and adhesives, Used in pesticides, cleaning agents, and rubber processing industries, Serves as a raw material in petrochemical industries for producing chemicals and synthetic products.

Light Aliphatic or Paraffinic Naphtha (LAN) mainly contains paraffinic and aliphatic hydrocarbons with straight-chain or branched structures. It has a lower aromatic content compared to HAN.

Applications

Used as fuel blending components in gasoline production, Used as feedstock in steam cracking to produce olefins such as ethylene and propylene, Used as cleaning solvents and in lighter industrial applications. **(Composition and applications of each 3.5 marks each for HAN & LAN)**

21. Describe the working principle of atmospheric and vacuum distillation. Compare their faction separated and applications. [U]/ [CO6]

Atmospheric distillation and vacuum distillation are important refining processes used to separate crude oil into different fractions based on their boiling points.

In atmospheric distillation, crude oil is first heated in a furnace to about 350–400°C and then introduced into a distillation column operating at atmospheric pressure. As the vapours rise through the column, components with different boiling points condense at different heights and are collected separately.

Fractions Separated

- Petroleum gas, Naphtha, Gasoline (petrol), Kerosene, Diesel, Heavy gas oil, Residue

Applications

- Production of fuels such as petrol, diesel, and kerosene.
- Provides feedstock for further refining processes like catalytic cracking and reforming.

Vacuum distillation is used for the heavy residue obtained from atmospheric distillation. The process is carried out under reduced pressure, which lowers the boiling points of heavy hydrocarbons and prevents thermal cracking or decomposition at high temperatures.

Fractions Separated

- Vacuum gas oil, Lubricating oil fractions, Paraffin wax, Asphalt or bitumen residue

Applications

- Production of lubricating oils, waxes, and bitumen.

- Provides feedstock for catalytic cracking and other secondary refining processes.

(Working principle 1.5 marks and 1 mark each for faction separated and applications for each distillation method)

22. Explain the procedure and significance of the penetration test for bitumen. [U]/[CO6]

The penetration test is a standard test used to determine the hardness or consistency of bitumen. It measures the depth to which a standard needle penetrates the bitumen sample under specified conditions.

Procedure: The bitumen sample is heated and poured into a standard container, then allowed to cool. The sample is maintained at a standard temperature of 25°C. A standard needle loaded with a weight of 100 g is placed just touching the surface of the bitumen. The needle is allowed to penetrate the sample freely for 5 seconds. The depth of penetration is measured in tenths of a millimetre (dmm). The average value from repeated tests is reported as the penetration value of the bitumen.

Significance of the Penetration Test

- It indicates the hardness or softness of bitumen.
- Helps in grading bitumen for different climatic and traffic conditions.
- Softer bitumen with higher penetration values is suitable for cold climates, while harder bitumen with lower penetration values is suitable for hot climates.
- Ensures uniform quality and proper selection of bitumen for road construction.
- Assists in predicting the performance and durability of pavement materials.

Thus, the penetration test is an important quality control test used to classify bitumen and determine its suitability for various engineering applications.

(Procedure and any three points of significance 3.5 marks each)